AURA-Twin

Automated Understanding & Real-time Analysis Twin

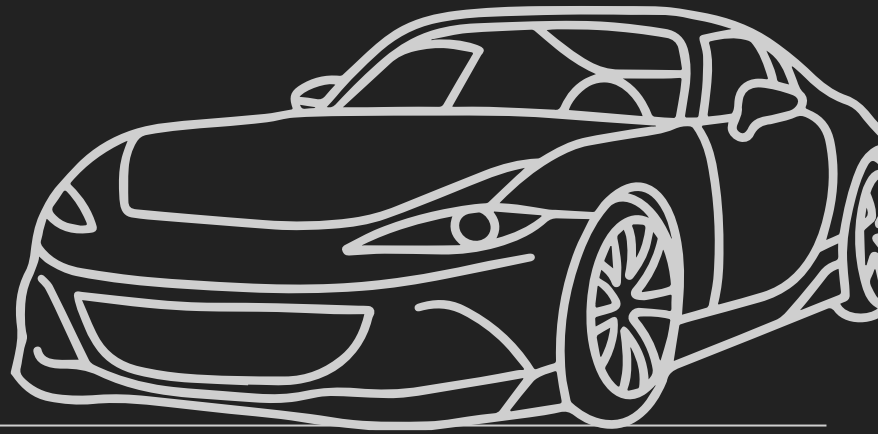
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Baseline Phase: Binary Engine Anomaly Detection (FordA)

Dataset & Scope

Utilized the benchmark **FordA dataset** for binary engine health monitoring, categorizing operations into **Normal** vs. **Faulty/Anomaly** based on 1D raw acoustic/vibration signals (500 time-steps per sample)

Model Architecture

Implemented a **Hybrid 1D-CNN + LSTM** deep learning architecture, combining 1D-CNN layers for local spatial/frequency feature extraction with LSTM layers to capture long-term temporal dependencies in vibration dynamics.

Results

High diagnostic accuracy—exceeding 95%—was achieved in test partitions using robust sequence learning.

Problem Statement & Scientific Challenges

Manual Feature Extraction Inefficiency:
Traditional signal methods (FFT/Wavelet) demand complex manual tuning and often miss early, non-linear degradation signatures.

Severe Industrial Noise Contamination:
Harsh mechanical interference and ambient noise heavily degrade the diagnostic accuracy of shallow learning models

Limitations of Binary Classification:
Binary outputs (Normal vs. Faulty) cannot identify the specific damaged component nor quantify defect severity levels.

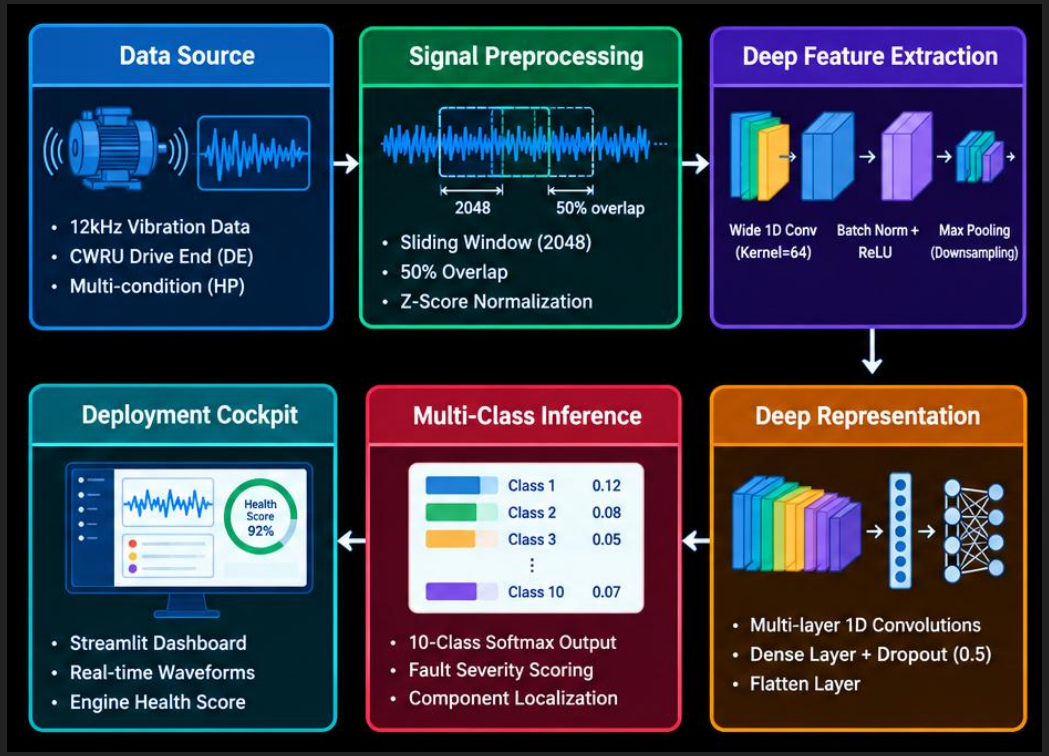
Objectives

- **Zero-Downtime Reliability:** Transitioning from reactive failure repairs to proactive, real-time health prognosis to eliminate sudden mechanical breakdowns and cut maintenance costs.
- **Precision Degradation Tracking:** Delivering early, sub-millimeter wear detection to accurately schedule component replacements before catastrophic failure occurs.
- **Autonomous Diagnostics:** Creating a self-contained intelligence pipeline that operates on raw operational data without requiring specialized human intervention.

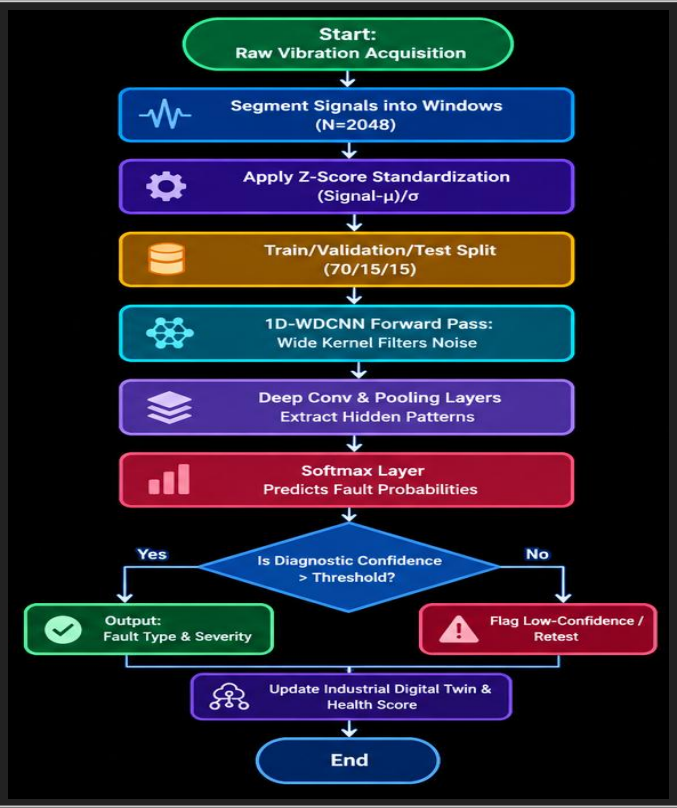
Proposed Solution

Building an end-to-end diagnostic system that ingests raw vibration signals, segments them using overlapping time windows (2,048 samples), and feeds them into a 1D Deep Convolutional Neural Network with a wide first-layer kernel to automatically suppress noise, extract fault signatures, and classify them across 10 distinct categories with real-time accuracy.

System Block Diagram



Execution Flowchart



Contents of this template

Paper Title	Year	Dataset Used	Training Approach	AI Model	Feature Extraction	Evaluation& Results
Vibration-based condition monitoring and fault diagnosis of rotating machinery: Fundamentals, signal processing techniques, standards, and emerging trends.	2026	Comprehensive Review Article; synthesizes data and findings from 238 peer-reviewed publications and international standards (e.g., ISO 10816, API 670).	Reviews Supervised Learning, Transfer Learning, Domain Adaptation, Unsupervised/Self-Supervised Learning, and Physics-Informed Hybrid Modeling	Covers Traditional Machine Learning (SVM, Random Forest), Deep Learning (1D-CNN, LSTM, Autoencoders), and Explainable AI (XAI).	Time-domain (RMS, Kurtosis), Frequency-domain (FFT), Time-frequency domain (STFT, Wavelets), and adaptive signal decomposition (VMD, EMD, Envelope Analysis).	Demonstrates superior diagnostic accuracy using adaptive decomposition and hybrid deep networks for non-stationary signals. Highlights Digital Twins, Physics-Informed AI, and Edge AI as key future directions for predictive maintenance.

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Paper Title	Year	Dataset Used	Training Approach	AI Model	Feature Extraction	Evaluation& Results
Bearing Fault Diagnosis Using CNN-LSTM Hybrid Model For Predictive Maintenance	2026	Case Western Reserve University (CWRU) Bearing Dataset (incorporating Drive End (DE) and Fan End (FE) vibration signals).	Data Split: Stratified split of 70% training, 15% validation, and 15% testing. Data Augmentation: Gaussian noise injection, random amplitude scaling, and random time-shifting. Optimization: Adam optimizer with Categorical Cross-Entropy loss over 20 epochs	Dual-branch Hybrid CNN-LSTM Architecture: Time-domain branch: Conv1D layers followed by an LSTM layer (64 units). Frequency-domain branch: STFT spectrogram conversion followed by Conv2D layers. Feature fusion via Dense layers and classification with a Softmax output layer.	Automated End-to-End Feature Extraction combining Time-Domain raw vibration signals and Time-Frequency representations (STFT) without hand-crafted features.	Accuracy: 96.00% (Training accuracy reached ~97%) Precision: 95.00% Recall: 94.00% F1-Score: 95.00% Comparative Performance: Outperformed KNN (71.34%), Decision Tree (95.69%), and ResNet-50 + SVM (95.51%).

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Predictive Maintenance Using Machine Learning for Engineering Systems Through Real-Time Sensor Data and Anomaly Detection Models	2026	The paper is a broad framework review and does not introduce a single primary dataset. However, it references standard time-series sensor data from IIoT environments (vibration, temperature, pressure, acoustics, current) used in various benchmark case studies (e.g., CNC machines, jet engines, wind turbines).	The paper surveys multiple paradigms: Supervised Learning (for labeled historical fault data), Unsupervised Learning (for unlabeled anomaly detection), Semi-Supervised/Self-Supervised (Autoencoders), Hybrid/Ensemble Learning, and Federated Learning. .	Supervised: Random Forest, XGBoost, SVM, Logistic Regression, KNN. Unsupervised/Anomaly Detection: Isolation Forest, K-Means, PCA. Deep Learning: Autoencoders, 1D CNN, LSTM, GRU, Hybrid CNN-LSTM.	Time-domain features (RMS, Kurtosis, Crest Factor), Frequency-domain features (Fast Fourier Transform - FFT), Spectral Entropy, and End-to-End automatic feature learning using 1D CNNs and LSTMs.	Discusses evaluation metrics such as Accuracy, F1-Score, AUC-ROC (for classification/anomaly detection) and RMSE (for Remaining Useful Life - RUL estimation). Summarizes that hybrid/deep models achieve high F1-scores and low false-positive rates across manufacturing, rail, aviation, and energy deployments.

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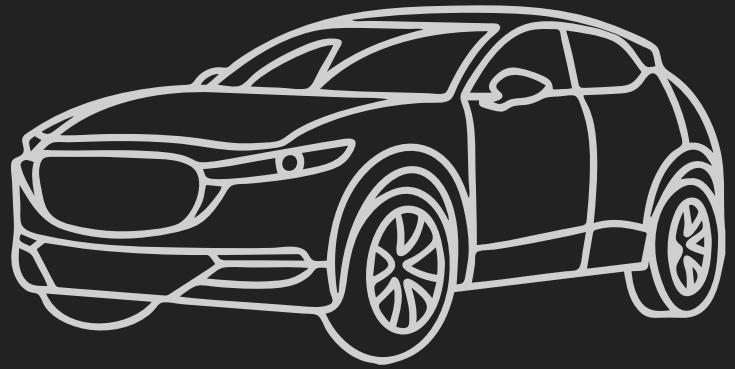
Paper Title	Year	Dataset Used	Training Approach	AI Model	Feature Extraction	Evaluation& Results
Trustworthy bearing fault diagnosis based on evidential deep learning and quadratic convolution neural network	2026	CWRU dataset (12 kHz drive-end bearing, 10 fault classes) contaminated with Gaussian white noise at 0 dB and -6 dB	Evidential Deep Learning (EDL) via Dirichlet distribution; improved loss function combining relaxed evidential loss, Fisher Information Matrix penalty, and KL divergence regularization; trained via SGD and ReLinear strategy	EQCNN (Evidential Quadratic Convolutional Neural Network built on Wide Deep CNN architecture)	Wide quadratic convolutional kernel (64 \times 1 \times 16) in Layer 1, embedded attention mechanism (Qttention), BN, ReLU, and Max Pooling across 6 QConv blocks	- Accuracy: 97.53%–99.91% at -6 dB SNR across motor loads, and 100% at 0 dB[- Uncertainty Quantification: Over 80% uncertainty for misclassifications and under 40% for correct predictions.Interpretability: Qttention maps accurately locate fault impact pulses

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Paper Title	Year	Dataset Used	Training Approach	AI Model	Feature Extraction	Evaluation& Results
Interpretable and Noise-Robust Bearing Fault Diagnosis for CNC Machine Tools via Adaptive Shapelet-Based Deep Learning Model	2026	CWRU (10 classes) and IMS run-to-failure datasets (4 classes) with Gaussian, Uniform, and Impulse noise (-6 dB to +8 dB)	Composite loss function (L_{CE} + \lambda_1 L_{cont} + \lambda_2 L_{cons}) combining cross-entropy, supervised contrastive loss (\lambda_1=0.05), and multi-scale consistency regularization (\lambda_2=0.02) with K-means++ initialization	Adaptive Shapelet-based CNN-BiLSTM	Multi-scale shapelets (lengths 16–256) with correlation-based distance, Gated parallel CNN (kernels 3, 5, 7), and Residual-enhanced BiLSTM	Accuracy: 87.67% (Gaussian), 85.65% (Uniform), and 82.33% (Impulse) at -6 dB on CWRU; 99.58% on IMS (clean) Superiority: Statistically significant improvement ($p < 0.01$) over standard baselines .Interpretability: FFT spectrum aligns with theoretical fault frequencies (e.g., BPFO = 107.4 Hz)

Data set Description

Utilizes the benchmark CWRU dataset (12 kHz Drive End vibration signals) encompassing 10 operating states: healthy baseline alongside 3 defect locations (Inner Race, Outer Race, Ball) across 3 severity levels (0.007”, 0.014”, 0.021”).



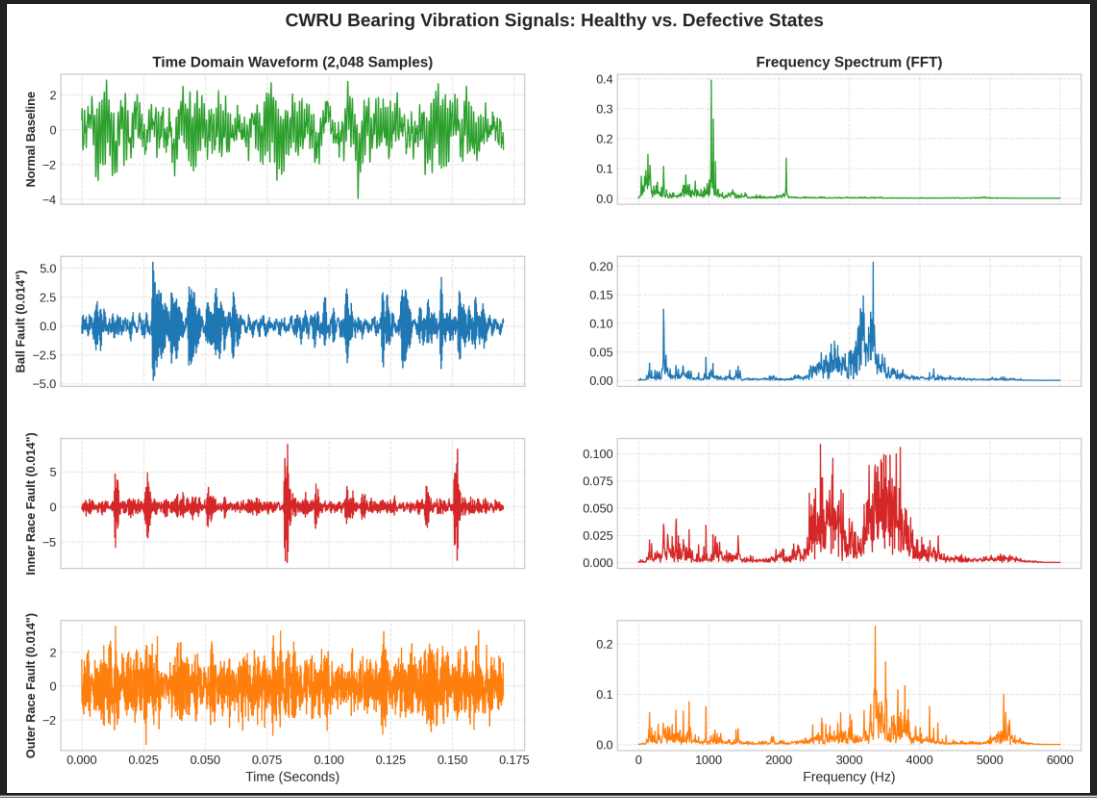
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Numerical Structure of CWRU 1D Vibration Data (Input Tensor Format)

Machine State (Condition)	t_1	t_2	t_3	t_4	t_5	...	t_2048	Label
Normal (Healthy)	0.5700	1.0608	1.2138	0.6451	-0.2296	...	-1.1505	0
Ball Fault (0.014")	0.0049	-0.6597	-0.3052	0.7326	0.3475	...	0.5332	1
Inner Race Fault (0.014")	-0.2597	0.0031	0.1007	-0.4654	0.1007	...	-0.3099	2
Outer Race Fault (0.014")	1.5442	-0.9408	0.1221	0.0666	-1.1298	...	-0.5651	3

Data set Description



Model Architecture

I will train everything using a baseline hybrid model combining CNN & LSTM, as well as another model combining CNN & ResNet. Additionally, I will experiment with a further model to achieve better accuracy.